

Python Iteration Practice Questions (25 Total)

Level 1 - Basic Iteration (For/While)

1. Print numbers from 1 to 10.
2. Print the sum of first N natural numbers (input N).
3. Print all even numbers from 1 to 100.
4. Print numbers from 10 down to 1 using a while loop.
5. Print the multiplication table of a given number up to 10.

Level 2 - Working with Collections

6. Calculate the sum of all elements in the list [2, 5, 8, 3, 10].
7. Count the number of vowels in a given string.
8. Find the maximum element in a list without using max().
9. Print each character of a string in reverse order using a loop.
10. Count how many even numbers are in a list.

Level 3 - Patterns and Nested Loops

11. Print a 4x4 square of stars using nested loops.
12. Print a right-angled triangle pattern of stars of height 5.
13. Print multiplication tables from 1 to 5 using nested loops.
14. Print all prime numbers between 2 and 50.
15. Implement FizzBuzz from 1 to 50 (print Fizz, Buzz, or FizzBuzz accordingly).

Level 4 - Intermediate/Complex Iteration Challenges

16. Check if a number is a palindrome (reads same forward and backward).
17. Print a pyramid of stars with n levels (center-aligned).
18. Given a number, print its binary representation without using bin().
19. Find the factorial of a number using a loop (not recursion).
20. Find all unique pairs in a list that sum to a given target value.
21. Print Fibonacci series up to N terms using iteration.

22. Reverse a list using only a loop (do not use reversed() or slicing).
23. Count the number of words in a sentence using loop only.
24. Create a histogram of character frequencies in a string using a loop.
25. Determine if a given string is a pangram (contains all letters A-Z).